

YOUNGJI KOH

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🌐 youngji-koh.github.io

SUMMARY

Researcher specializing in human–AI interaction, agentic and multi-agent systems, and multimodal computing for digital mental health and wellbeing.

RESEARCH INTEREST

Human–AI Interaction • Agentic and Multi-Agent Systems • Digital Mental Health • Multimodal Behavioral Modeling • LLM-based Contextual Reasoning • Ubiquitous Computing

EDUCATION

Ph.D. in School of Computing

Sep. 2021 – Aug. 2026

KAIST, Daejeon, Republic of Korea

Interactive Computing Laboratory (Advisor: Dr. Uichin Lee)

Dissertation: Multimodal Behavior Modeling and AI Agent Systems for Mental Healthcare in Intelligent Environments.

M.S. in School of Computing

Sep. 2019 – Aug. 2021

KAIST, Daejeon, Republic of Korea

Software Architecture Laboratory (Advisor: Dr. Sungwon Kang)

Dissertation: Deep Learning based Bug Report Summarization using Sentence Relations

B.S. in Computer Science and Engineering

Mar. 2015 – Aug. 2019

Ewha Womans University, Seoul, Republic of Korea

GPA: 4.02/4.30 (Summa Cum Laude)

RESEARCH EXPERIENCE

Visiting Researcher (Postdoctoral Fellow)

Sep. 2026 – Aug. 2027

Interactive Intelligent Systems Laboratory, The University of Tokyo (Host: Dr. Koji Yatani)

Fellowship: National Research Foundation of Korea (NRF) Postdoctoral Fellowship.

- Conducting research on human–AI interaction and AI-based mental healthcare systems.
- Investigating personalized and multi-agent AI approaches for supporting cognitive reframing, reflection, and mental wellbeing.

Undergraduate Research Intern

May 2018 – May 2019

Human-Computer Interaction Laboratory, Ewha Womans University (Advisor: Dr. Uran Oh)

- Built a touchscreen-based mobile application providing object-level verbal descriptions to support independent artwork exploration by people with visual impairments.
- Designed a mobile application for supporting conflict management in romantic relationships on mobile messengers.

FUNDED RESEARCH PROJECTS

Development of an Early Prediction System for Complex Diseases

Jun. 2025 – Aug. 2026

Institute for ICT Planning & Evaluation (IITP), Lead Researcher

- Designed a comorbidity-aware framework that coordinates disorder-specific agents for diagnosis and intervention planning.
- Focused on integrating LLM-based reasoning, multimodal sensing, and digital health applications for early intervention.

- **Keywords:** LLM Agent · Human-Centered Computing · Multimodal Sensing · Digital Health · Mental Health & Wellbeing

AI Models for Multimodal Mental Health Prediction

Sep. 2023 – Aug. 2025

LG Electronics, Lead Researcher

- Developed deep learning and machine learning models using IoT, mobile, and wearable data for depression and anxiety prediction.
- Evaluated detection accuracy and reliability through large-scale, real-world deployments.
- **Keywords:** Machine Learning · Mental Health & Wellbeing · Multimodal Sensing · Digital Health

IoT, Mobile, and Wearable Data Visualization for Mental Health Self-Reflection

May 2023 – Feb. 2025

LG Electronics, Lead Researcher

- Collected multimodal behavioral and physiological data via IoT, mobile, and wearable devices.
- Designed and evaluated personal informatics dashboards to promote self-reflection and wellbeing.
- The study was featured in national media outlets, including [Yonhap News](#), [Chosun Ilbo](#), [Herald Economy](#), and [Digital Times](#).
- **Keywords:** Data Visualization · IoT & Wearables · Mental Health · Ubiquitous Computing

Robotic Counseling System for Mental Health Support

Feb. 2023 – Oct. 2024

LG Electronics, Research Contributor

- Supported a two-week field study with 20 participants evaluating emotional awareness, engagement, and sense of place in robotic counseling.
- **Keywords:** Social Robots · Mental Health · Human-Robot Interaction · Digital Interventions

Context-Aware Multimodal Smart Speaker for Mental Health Self-Tracking

Sep. 2021 – Sep. 2023

LG Electronics, Lead Researcher

- Designed and developed a context-aware multimodal smart speaker for in-home mental health self-tracking.
- Led in-the-wild deployment for continuous behavioral monitoring and physiological data collection.
- The project received national media coverage highlighting its contribution to digital mental health innovation, including [Yonhap News](#), [Herald Economy](#), [Kookmin Ilbo](#), and [Maeil Business](#).
- **Keywords:** Smart Speaker · Context-Aware Sensing · Smart Home · Digital Health

PUBLICATIONS

Preprints

[J2] A Multimodal Sensor Fusion Approach for Mental Health Detection Using Mobile, Wearable, and IoT Sensors

Youngji Koh, Gyuna Kim, Chanhee Lee, Panyu Zhang, Yunhee Ku, Inhwon Choi, Jewoo Ryu, and Uichin Lee
Under review at *IEEE Journal of Biomedical and Health Informatics (J-BHI)*. IEEE. (Minor Revision Submitted)

Journal & Conference

[C8] Exploring Data-Driven Approaches to Stress Management: A Systematic Review of Stress Tracking, Intervention, and System Evaluation Methods

Youngji Koh, Gwangyoung Lee, Jeonghyun Kim, Yugyeong Jung, Hwajung Hong, and Uichin Lee
In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Barcelona, Spain, Apr. 13–17, 2026). CHI'26. ACM.

[C7] Leveraging Smartphone Human Interaction Routine Behavior Task Mining and Modeling for Daily Stress Monitoring

Hansoo Lee, Taehyeon Park, **Youngji Koh**, Jaegil Lee, and Uichin Lee
In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*. UbiComp'26. ACM.

[C6] Harnessing Home IoT for Self-tracking Emotional Wellbeing: Behavioral Patterns, Self-Reflection, and Privacy Concerns

Youngji Koh, Chanhee Lee, Eunki Joung, Hyunsoo Lee, and Uichin Lee

In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* (Espoo, Finland, Oct. 14–16, 2025). UbiComp'25. ACM.

[C5] 'In That Small Space with Just the Two of Us': User Experiences with Cumpa in a Robotic Counseling Center

Chanhee Lee*, Eunki Joung*, **Youngji Koh**, Esther Kim, Sohwi Son, Sunjung Kwon, and Uichin Lee (*: equal contribution)

In *Proceedings of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)* (Bergen, Norway, Oct. 18–22, 2025). CSCW'25. ACM.

[C4] Exploring Context-Aware Mental Health Self-Tracking Using Multimodal Smart Speakers in Home Environments

Jieun Lim*, **Youngji Koh***, Auk Kim, and Uichin Lee (*: equal contribution)

In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Honolulu, Hawaii, USA, May 11–16, 2024). CHI'24. ACM.

[C3] Interrupting for Microlearning: Understanding Perceptions and Interruptibility of Proactive Conversational Microlearning Services

Minyeong Kim*, Jiwook Lee*, **Youngji Koh**, Chanhee Lee, Uichin Lee, and Auk Kim (*: equal contribution)

In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Honolulu, Hawaii, USA, May 11–16, 2024). CHI'24. ACM.

[J1] Deep Learning-based Bug Report Summarization Using Sentence Significance Factors

Youngji Koh, Sungwon Kang, and Seonah Lee

Applied Sciences, 2022.

[C2] Bug Report Summarization using Believability Score and Text Ranking

Youngji Koh, Sungwon Kang, and Seonah Lee

International Conference on Artificial Intelligence in Information and Communication (ICAIIIC 2021). IEEE.

[C1] Deep Learning Based Bug Report Summarization Using Sentence Assessment Score

Youngji Koh and Sungwon Kang

Korea Software Congress (KSC 2020). *Best Paper Award*

Poster & Workshop

[W2] Designing Clinician-Centered Explanations for Sensing-based Longitudinal Mental Health Monitoring

Esther Kang, **Youngji Koh**, Seungwan Jin, and Uichin Lee

ACM UbiComp/ISWC 2026 XAI for U Workshop

[W1] Data Visualization for Mental Health Monitoring in Smart Home Environment: A Case Study

Youngji Koh, Chanhee Lee, Yunhee Ku, and Uichin Lee

IEEE VIS 2023 Workshop on Visual Analytics in Healthcare

[P4] LV-Linker: Supporting Fine-grained User Interaction Analyses by Linking Smartphone Log and Recorded Video Data

Hansoo Lee, Sangwook Lee, **Youngji Koh**, and Uichin Lee

ACM Symposium on User Interface Software and Technology (UIST 2022)

[P3] Supporting Object-level Exploration of Artworks by Touch for People with Visual Impairments

Nahyun Kwon, **Youngji Koh**, and Uran Oh

ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2019)

[P2] Analysis of Conflicts in Romantic Relationships on Mobile Messengers

Young-Ji Koh, Hyunggu Jung, and Uran Oh

HCI Korea 2019

[P1] **Real-time Fire Evacuation System based on Indoor Location Tracking and Route Optimization**
Young-Ji Koh, Naeun Go, ChaeYoon Kim, SooHee Lee, and Sangsoo Park
Korea Software Congress (KSC 2018). *Student Paper Competition, Merit Award*

PATENTS

International Patent Applications

[IP1] **Human Digital Twinning Method and System of Emotional Regulation for Emotional Labor in Workplaces Using Multi-Modal Sensor Data**
Uichin Lee, Duri Lee, Eunji Park, Youngji Koh, and Yunjo Hahn
U.S. Patent Application No. 18/212,435, filed Jun. 21, 2023; Publication No. US20240202623A1, published Jun. 20, 2024.

Domestic Patents

[DP3] **Apparatus for Collecting User's Experience According to the Situation and Method for Controlling Thereof**
Junhyung Kwon, Youngji Koh, Jieun Lim, and Uichin Lee
Korean Patent Application No. 10-2023-0060070, filed May 9, 2023; Publication No. 10-2024-0162897, published Nov. 18, 2024.

[DP2] **Human Digital Twinning Method and System of Emotional Regulation for Emotional Labor in Workplaces Using Multi-Modal Sensor Data**
Uichin Lee, Duri Lee, Eunji Park, Youngji Koh, and Yunjo Hahn
Korean Patent Application No. 10-2023-0038174, filed Mar. 23, 2023; Publication No. 10-2024-0092527, published Jun. 24, 2024.

[DP1] **System and Method for Providing Shortest Evacuation Route**
Sangsoo Park, Naeun Go, Youngji Koh, Chaeyoon Kim, and Soohee Lee
Korean Patent Application No. 10-2018-0162774, filed Dec. 17, 2018; Patent No. 10-2121672, granted Jun. 4, 2020.

HONORS AND AWARDS

Public Relations Award Donga Science, Korea	Dec. 2025
Creative Research Award K-Data Science Conference, Korea	Sep. 2025
Special Recognition for Outstanding Reviews ACM CHI	Dec. 2024
Ewha Womans University Scholarship Merit-based scholarship	Spring 2018

TEACHING EXPERIENCE

Teaching Assistant, KAIST CS492 Data Visualization	Fall 2021, Spring 2023
Teaching Assistant, KAIST CS204 Discrete Mathematics	Spring 2020

COMMUNITY SERVICE

HCI@KAIST

Student Steering Committee

2024, 2025

ACADEMIC SERVICE

Reviewer

ACM Transactions on Computing for Healthcare

IEEE Transactions on Computational Social Systems

ACM UbiComp/ISWC Notes and Briefs

ACM CHI